OptiSched: Uncertainty-Aware Enrollment Forecasting with CP-SAT Course Scheduling

Size every room to a calibrated demand bound, then schedule it conflict-free.

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−61% **unseated students** 153 → 60 seats vs point forecast, at 64.9% seat use

2/40 **courses overflow** vs 8 prior term, 9 point forecast

34/40 **inside 80% interval** 39/40 in 95% interval

3/3 **What-If re-solves optimal** 0 hard conflicts in every solve

1 Problem

Rooms are fixed months before registration. When demand outgrows a room, students start the term without a seat.

On the held-out term, a point forecast fell short on **19 of 40 courses**.

2 Approach

- **Forecast** Pooled ridge predicts next-term enrollment per course.
- **Calibrate** Split-conformal 80% and 95% intervals from backtest errors.
- **Size** Each course gets a room that fits its **80% upper bound**.
- **Schedule** CP-SAT assigns room, time slot and instructor under hard constraints.
- **Re-plan** What-If dry runs re-solve after a shock and list what moved.

3 Data: real vs. simulated

REAL UIUC registrar records (FOIA), 40 STEM courses · train on 19 terms, Fall 2016 to Fall 2025 · test on Spring 2026, held out · 2-year faculty habits

SIMULATED 16-room pool behind the OptiSched utilization numbers · preference weights 40/40/20 · What-If scenarios

Ten years of history predict one term ahead, not ten years out.

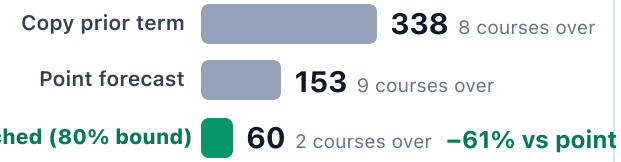
4 How it differs

	Prior term	Point forecast	OptiSched
Learns from history	1 term	19 terms	19 terms
Per-course interval	no	no	yes
Room buffer	flat +5%	flat +5%	80% bound

Standard timetabling tools treat enrollment as a fixed input. OptiSched schedules against a calibrated demand bound.

5 Results on held-out Spring 2026

STUDENTS OVER ROOM CAPACITY, LOWER IS BETTER



Baselines size rooms at forecast +5%.

FORECAST ERROR (MAE, SEATS)

28.25	18.03	17.24
copy prior term	seasonal naive	pooled ridge

−39% vs prior term, −4% vs seasonal naive. The interval, not point accuracy, drives the gain.

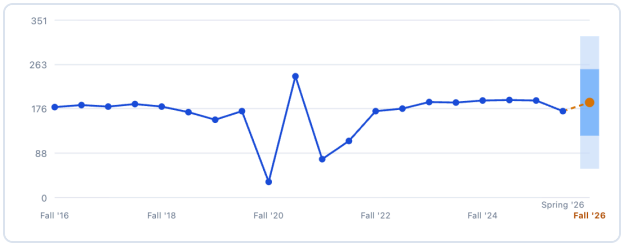
SCHEDULE QUALITY

	OptiSched	Actual UIUC
Seat utilization	64.9%	64.5%
Empty-seat waste	35.1%	36.6%
Faculty keep their usual slot	84%	n/a

Students. Fewer seat shortages in week one.

Administrators. Fewer room moves after registration.

6 Working demonstration

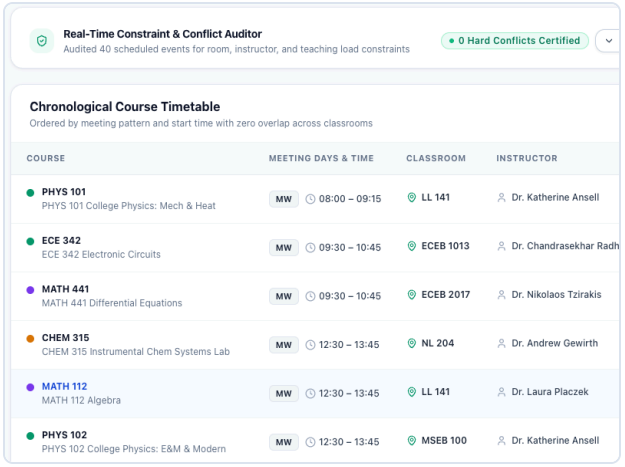


Live Fall 2026 forecast for STAT 400 with 80% and 95% intervals. Not part of the test.

WHAT-IF DRY RUNS, LIVE SCHEDULE UNTOUCHED

Enrollment surge	CS +20%, STAT +15%. 2 courses move to larger rooms.
Instructor unavailable	Only STAT 420 moves, to a qualified colleague.
Room capacity ×0.9	3 courses move to rooms that fit.

MASTER SCHEDULE



40 sections with room, time and instructor.

Code github.com/JoeDoan/CS5551_Advanced-Software-Engineering-

Reproduce `pytest backend/tests/test_uiuc_foia_replay.py`

FastAPI · OR-Tools CP-SAT · React + TypeScript · 353 automated tests

Extension Faculty Commercialization Advisor (UM-16C disclosures)